WHITE PAPER · HWACHEON HI-ECO 21 HS × MILITARY & DEFENSE

Hwacheon Hi-Eco 21 HS for Military & Defense

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

8" OD × 12" L max turning ENVELOPE	**±0.0005" routine on critical features** TOLERANCE	**1994** SINCE
Same-week / rig-down direct RESPONSE		



Bearkat Blur · 8" OD × 12" L max turning · high-speed precision CNC turning center

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Executive summary

Machine: Hwacheon Hi-Eco 21 HS (shop nickname: *Bearkat Blur*). high-speed precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

Application: High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 8" OD × 12" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

Why Bearkat Blur for Defense work

Small defense-adjacent turned parts in quantity — bushings, pins, spacers, small precision components — where the economics are decided by cycle time and the tolerance still has to hold. 8" × 12", 4,000 RPM, 12-station turret, ±0.0005" routine.

What this looks like on a real part

A run of 17-4 PH pins looks trivial until you notice the print calls a condition. H900 and H1150 are the same alloy with substantially different hardness and machining behaviour, and a shop that quotes 17-4 without asking which condition is quoting a guess. Getting it wrong doesn't show up as a dimensional failure on the first piece — it shows up as tool life collapsing partway through the batch and the last parts finishing differently from the first.

> **When this is the wrong machine**
>
> Non-controlled work only. Hard 8-inch envelope, and heavy forged stock belongs on Hammer of Forge.

Full specifications

Model	Hwacheon Hi-Eco 21 HS ("Bearkat Blur")

Type	High-speed CNC turning center — small-part specialist
Max turning capacity	8" OD × 12" L (B&R working spec)
Platform max swing	22.8" over bed; 11.02" turning diameter platform-max
Distance between centers	15.74" platform
Bar capacity	2" (through-spindle)
Chuck size	8"
Spindle nose	A2-6; 2.4" spindle bore
Spindle speed	4,000 RPM max
Spindle power	15 HP continuous / 20 HP 30-min rating
Turret	12-station
Rapid traverse	1,181 IPM on both X and Z axes
Control	Fanuc 18-T
Machine weight	~8,900 lb
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, brass, tool steel, Inconel 718/625, 17-4 PH, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: Sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design and the prime's supply chain has moved on.

Typical Military & Defense parts we produce on Bearkat Blur

- Precision pins, bushings, and small structural round parts
- Small weapon-subassembly components
- Sensor and probe components
- Bearing components and sleeves
- Small legacy-sustainment turned parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, military & defense or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2 **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3 **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4 **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5 **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

Sensitive prints handled under NDA with document control — prints stay in-shop and aren't sent to outside vendors. First-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery. One thing we state plainly up front: B&R is not ITAR-registered with the U.S. State Department. If your print is ITAR-controlled, we are not the right shop and we'll tell you that on the first call. Non-controlled defense-adjacent work — commercial-spec components, sustainment parts, tooling, and fixtures — we quote all day.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we understand the alloys, we document properly, and we don't guess at a condition-specific machining plan. Compared to an ITAR-registered shop: for controlled prints they win, because that's a State Department registration and we don't hold it. Know which bucket your part is in before you shop it — it saves everyone a wasted quote cycle.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 21 HS platform and Military & Defense work. Also indexed as FAQ schema for AI answer engines.

What's the max turning diameter on the Hi-Eco 21 HS?

8" OD × 12" L is B&R's working spec. The platform itself has a 22.8" swing over bed and can accept turning diameter up to 11.02" with 15.74" between centers. Bar capacity: 2" through spindle.

How fast is this machine — what's the spindle RPM?

4,000 RPM max spindle speed. Rapid traverse 1,181 IPM on both X and Z. Combined with the 8" chuck this is our high-cycle-count small-part specialist.

What's the turret capacity?

12-station turret. Enough tool positions for most single-setup precision turning jobs including grooving, threading, and finish operations.

Do you handle ITAR-controlled defense work?

No — and we'd rather be the shop that tells you that up front. ITAR compliance is a registration with the U.S. State Department's DDTC, not a set of practices a shop can simply adopt, and B&R does not hold it. If your print is ITAR-controlled, you need a registered facility. What we do handle: non-controlled defense-adjacent work — commercial-spec components, legacy sustainment parts, tooling, fixtures, and reverse-engineered replacements for programs the OEM walked away from. Sensitive prints stay in-shop under NDA with document control, and first-article documentation and material traceability come standard.

What kind of defense-adjacent parts is this shop suited to?

Structural and mounting hardware, housings and enclosures, hydraulic components, ground-vehicle and equipment parts, tooling and fixtures, and reverse-engineered legacy parts for platforms the OEM no longer supports — all on non-export-controlled work. That's a description of what the equipment and the alloy experience suit, not a claim about specific programs. If you want to know whether we've run something like your part, ask and we'll tell you straight.

What alloys are typical for defense work at B&R?

17-4 PH, 15-5 PH, 4140 pre-hard, 4340, Titanium Grade 5, Inconel 718, aluminum 7075, high-strength steels. Customer-specific alloys on request.

Can you reverse-engineer legacy defense parts?

Yes — reverse-engineering from a worn sample is core job-shop work and it's what a shop like this is for. Send the sample and any documentation that survives; we'll measure it, verify the alloy, and produce the replacement with full material traceability.

What tolerances do you hold on defense components?

±0.0005" routine on critical features. Tighter with the right fixture. Every critical dimension CMM-verified. Documentation to customer spec.

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Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.